(1) Murphy 1.2 Solve using matrices.

(2) Murphy 1.15

(3) Murphy 1.17

(4) The isobaric heat capacity $C_P(T,P)$ is a function of temperature and pressure. The ideal gas isobaric heat capacity is the low pressure limit, such that, $C_P(T) = \lim_{P\to 0} C_P(T,P)$. Data is supplied for $C_P(T)/R$ versus $T(K)/1000$. Here R is the ideal gas constant so that $C_P(T)/R$ is dimensionless, and $T(K)$ is the temperature in Kelvin. Define $t = T(K)/1000$. These are kiloKelvin units, so to speak.

(a) Use least squares to fit the data to a third-order polynomial

$$\frac{C_P(T)}{R} = A + Bt + Ct^2 + Dt^3$$

where A, B, C, and D are adjustable constants that you will determine by fitting the data for molecular oxygen (O_2) in the gas state, which is given to the right. Assume that a pressure 0.01 atm is low enough to give ideal gas heat capacity values.

(b) Compare your results with fits for $C_P(T)$ given in the following sources which can be found on the course Blackboard webpage:

(i) Appendix B.17 of Murphy,
(ii) Table A-21 of Moran, Shapiro, Boettner, and Bailey
(iii) Appendix B.2 of Felder and Rousseau,
(iv) Appendix C of Smith, Van Ness, and Abbott, and
(v) The NIST Chemistry Webbook

Be sure to plot your results as for $C_P(T)/R$ versus $T(K)/1000$. This will require adjusting the fits from the sources above. Comment on the quality of the fits compared to the data.

T(K)/1000	Cp/R (0.01 atm)
0.3	3.5345
0.35	3.5717
0.4	3.6212
0.45	3.6787
0.5	3.7396
0.55	3.8008
0.6	3.8599
0.65	3.9155
0.7	3.9672
0.75	4.0145
0.8	4.0577
0.85	4.0970
0.9	4.1327
0.95	4.1652
1	4.1948
1.05	4.2219
1.1	4.2469
1.15	4.2698
1.2	4.2912
1.25	4.3112
1.3	4.3300
1.35	4.3479
1.4	4.3651
1.45	4.3815
1.5	4.3975

(5) A tank can be filled with liquid water from three different water supply lines which are labeled A, B, and C. The flow rates from each supply line is constant. The volume of the tank is unknown. When supply lines A and B are used simultaneously to fill the tank, the filling process takes half an hour. Similarly, when A and C are used, it takes three-quarters of an hour. And when B and C are used, it takes one hour.

(a) How long does it take to fill the tank when all three supply lines are used simultaneously?

(b) How long does it take for each of the supply lines alone to fill the tank?

1) $C_3H_5(NO_3)_3 \longrightarrow CO_2 + H_2O + N_2 + O_2$

$\quad a \qquad\qquad\qquad b \qquad c \qquad d \qquad e$

C	$3a$	$=$	$1b + 0c + 0d + 0e$ ✓
H	$5a$	$=$	$0b + 2c + 0d + 0e$ ✓
O	$9a$	$=$	$2b + 1c + 0d + 2e$ ✓
N	$3a$	$=$	$0b + 0c + 2d + 0e$ ✓

$$\text{rref}\begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 2 & 0 & 0 & 5 \\ 2 & 1 & 0 & 2 & 9 \\ 0 & 0 & 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 5/2 \\ 0 & 0 & 1 & 0 & 3/2 \\ 0 & 0 & 0 & 1 & 1/4 \end{bmatrix} \times 4$$

$b = 3$
$c = 5$
$d = 3$
$e = 1$

$a = 1 \times 4 = 4$

$$\boxed{4\, C_3H_5(NO_3)_3 \longrightarrow 3CO_2 + 5H_2O + 3N_2 + 1O_2}$$

Have to notice that there is 5 unknowns but only 4 eqns, set one of the unknowns =1 and then solve. Half-credit, -7.5 pts

7.5/15

2) $HNO_3 + CH_3OH \longrightarrow C_3H_7NO_2 + CO_2 + H_2O$

	a		b	c	d	e
C	0	=	-1	3	1	0
H	1	=	-4	7	0	2
N	1	=	-0	1	0	0
O	3	=	-1	2	2	1

$$\text{rref}\begin{bmatrix} -1 & 3 & 1 & 0 & 0 \\ -4 & 7 & 0 & 2 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 2 & 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & & & & 10/3 \\ & 1 & & & 1 \\ & & 1 & & 1/3 \\ & & & 1 & 11/3 \end{bmatrix} \times 3$$

b = 10
c = 3
d = 1
e = 11

a = 1 × 3

$$\boxed{3HNO_3 + 10CH_3OH \longrightarrow 3C_3H_3NO_2 + 1CO_2 + 11H_2O}$$

Grams of Methanol req to reduce nitrant content to 10L of well water from 54 mg/L to 10mg/L

$$54\frac{g}{mL} - 10\frac{g}{mL} = \frac{44\,mg}{L} \times \frac{1g}{10^3 mg} = 0.044\,g/L$$

$$\frac{0.044\,g}{L} \text{ of } HNO_3 \times \frac{1\,mol\ HNO_3}{63g} \times 10\,L = 0.0070\text{ mol } HNO_3$$

$$0.0070\,mol\ HNO_3 \times \frac{10\,mol\ CH_3OH}{3\,mol\ HNO_3} \times \frac{32\,g\ CH_3OH}{1\,mol\ CH_3OH} = \boxed{0.75\,g\ CH_3OH}$$

20/20

3) 1% to 8% by weight of hydrogen

wt % of H in C_2H_5OH, CH_4, H_2O, $C_6H_{12}O_6$

$$C_2H_5OH = \frac{6(1.0079)}{2(12.01)+6(1.0079)+1(15.99)} = 13 \text{ H wt \%}$$

$$CH_4 = \frac{4(1.0079)}{4(1.0079)+1(12.01} = 25 \text{ H wt \%}$$

$$H_2O = \frac{2(1.0079)}{1(15.99)} = 11 \text{ H wt \%}$$

$$C_6H_{12}O_6 = \frac{12(1.0079)}{6(12.01)+12(1.0079)+6(15.99)} = 7 \text{ H wt \%}$$

From the calculated wt % of hydrogen most were above the range of 8%. Which support the idea that it could be 50% wt.

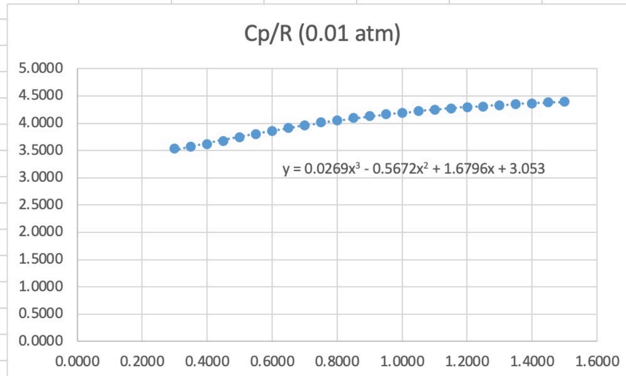
-7.5 pts, incorrect answers and conclusion but made a solid effort

7.5/15

4)

T(K)/1000	Cp/R (0.01 atm)
0.3000	3.5345
0.3500	3.5717
0.4000	3.6212
0.4500	3.6787
0.5000	3.7396
0.5500	3.8008
0.6000	3.8599
0.6500	3.9155
0.7000	3.9672
0.7500	4.0145
0.8000	4.0577
0.8500	4.0970
0.9000	4.1327
0.9500	4.1652
1.0000	4.1948
1.0500	4.2219
1.1000	4.2469
1.1500	4.2698
1.2000	4.2912
1.2500	4.3112
1.3000	4.3300
1.3500	4.3479
1.4000	4.3651
1.4500	4.3815
1.5000	4.3975

$$y = 3.053 + 1.6796x - 0.5672x^2 + 0.0269x^3$$ ✓



After taking a look at the data from the tables the values that were generated by the line of best fits that fits the specification of having a cubic term and polynomial behavior. Really coincide with the values from the tables in black board for being in the same ranges for example. The A value for the line of best fit was almost the same for all table ranging by just a couple of decimal point off as the end of the value.

Should have graphed other data and explained in detail, -10 pts

15/25

5)

$\cancel{A+B = \frac{1}{2}\,hr}$
$\cancel{A+C = \frac{3}{4}\,hr}$
$\cancel{B+C = 1\,hr}$ $\longrightarrow$

$$\frac{1}{2}(A+B) = V \qquad \frac{3}{4}(A+C) = V \qquad B+C = V$$

$$\text{rref}\ \begin{array}{ccc} A & B & C \\ \end{array}$$
$$\text{rref}\left[\begin{array}{ccc|c} 1/2 & 1/2 & 0 & 1 \\ 3/4 & 0 & 3/4 & 1 \\ 0 & 1 & 1 & 1 \end{array}\right] = \begin{array}{cccc} 1 & 0 & 0 & 7/6 \\ 0 & 1 & 0 & 5/6 \\ 0 & 0 & 1 & 1/6 \end{array}$$

Flow Rates for different supply lines connected
- ↳ A = 7/6 tank/hr
- ↳ B = 5/6 tank/hr
- ↳ C = 1/6 tank/hr

A) All 3 used at same time.

x in time

$$\rightarrow x\left(\frac{7}{6}\frac{tank}{hr} + \frac{5}{6}\frac{tank}{hr} + \frac{1}{6}\frac{tank}{hr}\right) = 1\,tank$$
$$x\left(\frac{13}{6}\frac{tank}{hr}\right) = 1$$
$$x = \frac{6}{13}\,hr = 27.69\,min$$ ✓

25/25

B)
$$x\left(\frac{7}{6}\frac{tank}{hr}\right) = V \qquad x = \frac{6}{7}\,hr = 51.43\,min$$ ✓
$$x\left(\frac{5}{6}\frac{tank}{hr}\right) = V \qquad x = \frac{6}{5}\,hr = 72.00\,min$$ ✓
$$x\left(\frac{1}{6}\frac{tank}{hr}\right) = V \qquad x = 6\,hr = 360\,min$$ ✓

supply line C takes the longest to fill tank.